

Event Density

A Generative Substrate Ontology for Cross-Scale Physics

Whitepaper — Program Overview, Architecture, and Falsification Path

Wave-3 generative document. Public-facing, non-technical orientation layer. Standalone, citable, repo-top-level document.

Executive Summary

Event Density (ED) is a research program that asks a single foundational question: *can the diverse phenomena of physics — quantum mechanics, gauge theory, gravity, black holes, turbulence, decoherence, entanglement — be generated from a small, well-specified substrate ontology?*

The program’s answer is conditional: **yes, given thirteen posited primitives and a kernel calculus built from two substrate-level memory kernels**. The thirteen primitives are listed in a position paper and form a minimal generative ontology. The two kernels — designated V1 (a finite-width retarded kernel) and V5 (a finite-memory cross-chain correlation kernel) — supply the dynamical content from which physical structure is derived.

The substrate→continuum bridge used across every continuum-level result in the corpus is the **Diffusion Coarse-Graining Theorem** (DCGT, Paper_073), at verdict tier **M3 within an A→regime hydrodynamic-window scale-separation assumption** ($\ell_{\text{ED}} \ll R_{\text{cg}} \ll L_{\text{flow}}$). Continuum equation forms (diffusion operator, propagator hierarchy, memory-kernel, KMS-thermal) are FORM-FORCED from substrate kernel structure (V1, V5) and substrate-graph averaging; numerical transport coefficients (diffusion constant, Maxwell relaxation time, etc.) are VALUE-INHERITED via empirical matching; one cross-domain instantiation (Lieb–Robinson velocity bound) carries an A→dimensional-analogy label rather than full derivation; explicit breakdown conditions (strong-gradient, near-substrate-scale, near-singularity, strong-curvature regimes) are predicted, not papered over. Because DCGT is upstream of every continuum-level result, its M3-with-regime-caveat status is inherited by every downstream continuum prediction.

What ED claims is *cross-domain reach plus methodological consistency*. Across eight **structurally complete research arcs with declared open derivations** (quantum kinematics, form-level quantum field theory, gravity, black holes, quantum computing limits, entanglement, soft matter / Navier–Stokes, foundational kernel theory), ED produces results that are **form-forced** by the primitives and an honestly-disclosed set of paper-specific postulates. Numerical values are inherited from empirical matching rather than derived from first principles. This split — *form forced, values inherited* — is the program’s methodological signature. “Structurally complete” means each arc has an integrated framework with explicit falsification criteria and a registry of what is form-FORCED, value-INHERITED, or OPEN; it does **not** mean each arc contains no open derivations. Open derivations are named (Route A, Phase-3 GR, YM mass-gap mechanism, NS smoothness at Intermediate Path C) rather than hidden.

The program’s most ambitious recent synthesis is the **Substrate–Cosmology Boundary Unification (SCBU)** and its six-paper extension arc, the **ED-SC 4.x series**. SCBU is offered as the framework’s **organizing structural claim**: that the black-hole horizon, the cosmic decoupling surface, the deep-MOND galactic asymptote, the quantum-computing platform-coherence threshold, the soft-matter hydrodynamic Q-factor, and the substrate-scale fixed-point parameter are projections of a single substrate–cosmology boundary onto distinct parameter

axes. ED-SC 4.x organizes these projections within a unified four-regime cross-scale model (ultraviolet / transition / infrared / cosmological).

Honest framing of SCBU’s current status (post-2026-05-17 Route A closure). SCBU is now an **M3-supported structural claim at substrate-parameter-INHERITED level** (Paper_027-analog). Route A — substrate-derived $\ell_{V_5}(H_0)$ relating V5 bandwidth to the cosmological rate — **closed at substrate-parameter-INHERITED level** via the Route A audit cascade (multi-route convergence audit option (ii); closure documented in Paper_ED_SC_4.2 §1.5). The closure delivers $H_0 = (8\pi/3)^{1/2} \ell_{PC}^{1/2} \Theta_{ED}^{1/2}$ with $\Theta_{ED}^{\text{Planck-units}} \approx 10^{-122}$ substrate-parameter-INHERITED at P12 primitive level (analogous to Paper_027’s inheritance of $\ell_P, \hbar, c$). Cross-arc consequences: **Cosmology Arc M3 Trinity** (Cos_01 + Cos_05 + Paper_038_5 all at M3 via the same Q1A + Q2A + C3 + IC-INHERITED chain + Route A4 substrate-parameter-INHERITED closure); **ED-SC 4.x arc-wide M3 upgrade** across all six projections. The six-projection picture is now a substrate-graph-derived structural claim at substrate-parameter-INHERITED level. Tightening work — per-projection exponent derivation; substrate-graph derivation of $\Theta_{ED} + \tau_{V_5}$ values from sub-primitive content (analog of longstanding “what determines $\ell_P, \hbar, c$ values?”) — remains as substrate-research-frontier work, **not load-bearing** for the M3 verdicts.

The post-pivot gravity line and the substrate-evaluation wave (June 2026). Two developments postdate the eight-arc structure. A six-paper **gravity line** (GR-I ... GR-IV, KM-I/II, the One-Field capstone) derives — from the substrate’s irreversible arrow placed *in the law* — the weak-field Einstein metric with its **factor-of-two light bending**, the **khronometric** gravitational class (GW-clean, Lorentz-safe), the preferred-frame parameter α_1 **suppressed below the experimental bounds by more than 70 orders of magnitude**, and the MOND / dark-energy phenomenology, unified as *one field with one scale* (the full nonlinear Einstein equation remains open). Separately, a **substrate-evaluation wave** reports *build-and-run* results from the certified rule — including the program’s only *proven* negative (the finite-memory prime ceiling) and its reach statement, *Form and Flesh: The Two Walls*. Both are developed in §5.

The program is **falsifiable**. Each result carries explicit falsification criteria. The SCBU framework predicts synchronized co-variation of the six projections under Hubble-tension-class shifts of H_0 ; with Route A now closed at substrate-parameter-INHERITED level, **per-projection scaling exponents are derivable in principle via the Route A inheritance chain** (a tightening-work direction, not a load-bearing OPEN). $a_0 \propto H_0^1$ is structural (Paper 029); $R_H \propto H_0^{-1}$ is structural (Paper 028); $\xi_{\text{canonical}} \propto H_0$ at leading order via Route A4; the remaining per-projection exponents ($\mathcal{M}_{\text{crit}}, Q$) sharpen with per-projection derivation work. The sharpest currently-clean falsifier in the corpus remains the **BTFR slope-4 with zero intrinsic scatter** (Paper 031), testable against galaxy-rotation-curve catalogs at present precision.

This whitepaper is the front door. It explains what ED is, what it claims, what it does not claim, and how to navigate the 100+ research papers that constitute the program (the numbered Paper_NNN series plus theorem-stub additions T19–T21, the cross-arc synthesis Paper_SCBU, and the six-paper ED-SC 4.x extension arc).

§1 — Motivation

Physics, considered as a single intellectual artifact, is a patchwork. Quantum mechanics and general relativity remain unreconciled in their foundations. Standard-model gauge theory works at one scale; cosmological dynamics works at another. The black-hole information paradox lives in a third register; turbulence and condensed-matter rheology in a fourth. Each domain has its

own framework, its own postulates, and its own community.

A generative substrate ontology proposes an alternative architecture: *rather than treating each domain as autonomous, derive each domain's structural content from a common substrate*. The substrate is posited, not derived. From the substrate, dynamical kernels are built. From the kernels, coarse-grained continuum theories emerge. From the continuum theories, specific physical predictions follow.

ED's intellectual neighborhood is the **substrate-ontology research program**: 't Hooft's cellular-automaton interpretation of QM, Wolfram's Ruliad / hypergraph-rewriting program, and the causal-set program (Sorkin et al.). These are programs that posit substrate ontologies and exhibit downstream physics; they are not axiomatic reconstructions in the sense of Hardy (2001), CDP (2011), or Masanes-Müller (2011), where finite-dimensional QM is derived from operational axioms by closed proofs. ED should be read against the substrate-ontology lineage, not against the operational-reconstruction lineage. The methodological discipline ED adds to its lineage is the form-FORCED / value-INHERITED split (Paper 095) and per-paper audit tables with explicit P/D/A/I labels — improvements on the field's norm, not equivalents of closed-proof reconstructions.

The motivation for ED is not that the patchwork is wrong — each domain's predictions are excellent within their regimes — but that the patchwork may hide **cross-domain structure**. If a single substrate-level object underlies phenomena across the black-hole horizon, the cosmic decoupling surface, the substrate-scale fixed-point, the deep-MOND galactic acceleration, the matter-wave quantum-classical boundary, *and* the Navier–Stokes Q-factor, the cross-domain structure would itself be a physical fact worth identifying. SCBU (§7) offers this as a six-projection picture of a single substrate-cosmology boundary; **post-2026-05-17 Route A closure, the picture is M3-supported at substrate-parameter-INHERITED level** (see Cosmology Arc M3 trinity + ED-SC 4.x arc-wide M3 upgrade per the Route A closure documented in Paper_ED_SC_4.2 §1.5). The program is honest about this status throughout, including the substrate-parameter-INHERITED status of Θ_{ED} and τ_{V_5} (analogous to Paper_027's inheritance of $\ell_P, \hbar, c$).

ED's role is to make this cross-domain structure explicit, falsifiable, and structurally honest. The program does not replace standard physics operationally; it supplements it with a substrate-level account of why the standard results take the forms they do.

§2 — The 13 Primitives

ED's substrate is specified by thirteen primitives. They are **posited** — not derived from anything deeper. The enumeration below is the canonical list per `position-paper/paper_ED_Framework_13_Primitives` and the audit document `primitives/PRIMITIVE_LOAD_BEARING_AUDIT.md`; downstream papers reference these P-numbers.

A note on minimality. The 13 are claimed *necessary-for-current-derivations* (removing any one blocks at least one downstream derivation, audited in `PRIMITIVE_LOAD_BEARING_AUDIT.md`). They are **not** claimed minimal in an absolute sense: smaller primitive sets generating the same downstream content may exist. We do not claim otherwise. P03 in particular bundles channel indexing, locus indexing, and spatial homogeneity; this is a working consolidation, not a forced reduction. The arrow of time is supplied jointly by P11 + P13 + V1's retardation property (Theorem T18), and the redundancy question among these three is structurally open.

The thirteen primitives, in plain language:

P01 — Event-density layer existence. A pre-quantum substrate exists as a primitive structural layer. The substrate is not a Hilbert space, not a smooth manifold, not a field theory — it is a discrete graph-like structure equipped with primitive scalar, relational, and angular quantities. P01 is the foundational ontological-existence commitment that grounds every paper in the program.

P02 — Participation as primitive relation. Chains (persistent micro-event sequences) participate in channels as a primitive substrate-level relation. Participation is the substrate’s mechanism for chain-channel interaction; without P02 no participation measure can be defined.

P03 — Channel + locus indexing; spatial homogeneity. The substrate supplies a discrete index set of channels $\mathcal{K}$ and a (discrete or continuous) index set of loci, with primitive translation-invariance: shifting all loci by a common displacement leaves the participation-graph adjacency structure intact. This primitive bundles three commitments (channel index, locus index, spatial homogeneity); they could in principle be split.

P04 — Bandwidth (non-negative additive scalar, four-band partition). Each channel K at each locus u carries a real-valued non-negative quantity $b_K(u) \geq 0$, additive under channel decomposition. Bandwidth further decomposes into four mutually orthogonal substrate-level bands: internal, adjacency, environmental, commitment-reserve.

P05 — Polarity-transport between adjacent loci. The substrate-level connection structure — the mechanism by which polarity at one locus is transported to a neighboring locus along edges of the participation graph. P05 is the substrate-level origin of phase, gauge connection, and parallel-transport phenomena.

P06 — Spatial dimension $D = 3 + 1$. The substrate’s spatial axis is $\mathbb{R}^3$, plus one time dimension. Consistent with multiple independent derivations elsewhere in the corpus (PDE-uniqueness axioms, spinor-bundle structure, acoustic-metric signature).

P07 — Channel structure as ontological primitive. Channels are structurally distinguishable carriers of substrate content, with identities intrinsic to the participation graph rather than derived from external organizational choice.

P08 — Substrate scale ℓ_{ED} . The substrate has a characteristic edge length ℓ_{ED} . **The identification $\ell_{\text{ED}} \equiv \ell_P$ (the Planck length) is a value-inheritance step**, fixed by Newton-recovery: matching the substrate-derived form $G = c^3 \ell_{\text{ED}}^2 / \hbar$ to the empirical value of Newton’s constant determines ℓ_{ED} to be ℓ_P . This is not a first-principles derivation of G ’s numerical value; it is the substrate-side identification of the length-scale anchor (see §3 + §7 for honest framing).

P09 — $U(1)$ -valued polarity. Each channel carries an angular variable $\pi_K(u) \in U(1) \cong S^1$, the substrate’s unique angular primitive. Supplies the angular / phase structure that propagates to gauge fields and quantum phase.

P10 — Rule-type primitive. The substrate supports multiple structurally distinct rule-types $\tau_\bullet$, each with its own participation measure. Matter and gauge are *different rule-types*, not different states of the same one. The V1 and V5 kernels (see §3) are rule-types.

P11 — Commitment with environmental phase-randomization, irreversible. Discrete substrate-level events at which a chain’s multi-channel participation collapses to a single channel, with phase-randomization on a uniform $U(1)$ distribution. Commitment is irreversible — no substrate-level dynamics maps a post-commitment state back to its pre-commitment superposition. P11 supplies the directional content of time and the retardation property of the V1 kernel.

P12 — Stability landscape. Substrate-level functional $\Sigma = \text{Coh} - \text{Str} - \text{Grad}$ whose extrema

govern chain dynamics. Carries the substrate-level coherence, strain, and gradient content for gravitational physics.

P13 — Time homogeneity; primitive event-discreteness. The substrate’s dynamical primitive is invariant under time translation; the substrate has no preferred absolute time origin. Events on the substrate are primitively discrete (the manifold is not infinitely refinable). P13 supplies H_0 as a cosmological-rate parameter.

A fourteenth slot, **P14**, is reserved as a placeholder for bilocal strain coupling. P14 is *not* currently in the canonical 13; its presence as a reserved slot indicates that the count is the working consolidation, not a load-bearing claim. Whether P14 should be elevated, or whether its content can be derived from the existing 13, is structurally open.

Relationship to the primitives/ directory. The **primitives/** directory is now organized in two tiers: - **Top-level files P01_*.md through P13_*.md** — the canonical primitive reference cards. These match the enumeration above 1-to-1; downstream papers cite them by P-number. - **primitives/concepts/ subdirectory** — thirteen process-level treatments of the *underlying ontological concepts* the canonical primitives organize (micro-event, chain, participation, participation_bandwidth, event_density, ed_gradient, channel, multiplicity, tension_polarity, individuation, commitment, thickening, relational_timing). The mapping to canonical primitives is *not 1-to-1*: chains and individuation are derived rather than canonical-primitive; micro-event and event-density jointly contribute to P01; P05 polarity-transport and P06 spatial dimension have no concept-file counterpart.

See **primitives/README.md** for the full two-tier structure and citation convention.

The thirteen primitives are **the** ontological commitment of the ED program. Every result in the corpus rests on them — though as noted, “minimal” means *necessary-for-current-derivations*, not *absolutely minimal*.

§3 — Kernel Calculus (V1 / V5)

ED’s substrate dynamics are carried by two substrate kernel rule-types (in the sense of P10). Both are finite — neither requires infinite-resolution data — and both are forward-causal under P11.

V1 — Finite-width retarded vacuum kernel. V1 propagates substrate amplitudes from prior to later times with a characteristic width $\ell_{V1} \sim \ell_P$ and a retardation property forced by P11. V1 supplies:

- The substrate-level propagator entering quantum field theory.
- The UV regulator at the Planck scale (preventing trans-Planckian divergences).
- The kinematic viscosity scale entering hydrodynamics ($\nu \sim \ell_{V1}^2 / \tau_{V1}$).
- The friction coefficient entering the Universal Mobility Law.
- The propagation rate identified as substrate c (constant under coarse-graining).

V5 — Cross-chain finite-memory correlation kernel. V5 supplies cross-chain correlations with finite bandwidth ℓ_{V5} and finite memory τ_{V5} . V5 carries the cross-chain content that V1 (within-chain) does not. V5 supplies:

- Maxwell-form stress-relaxation in viscoelastic materials ($\tau_M \sim \tau_{V5}$).
- The substrate origin of bipartite entanglement budget partitioning (monogamy).
- The cross-chain bandwidth that caps Q-Compute multiplicity.
- The decoupling-surface mechanism at black-hole horizons ($\Gamma_{\text{cross}} \rightarrow 0$).
- The Page-curve mechanism for Hawking radiation entanglement.

- The cutoff $\omega_c = c/\ell_P$ resolving the trans-Planckian problem.

The two kernels together form a **kernel calculus**: a small, complete set of substrate-level objects from which coarse-grained continuum theories emerge under the Diffusion Coarse-Graining Theorem (DCGT).

Why only two kernels? A central structural claim of ED is that V1 + V5 are sufficient. Higher-order kernels are admissible in principle but, across all current research arcs, V1 and V5 suffice. The same V5 **rule-type** appears with structurally identical roles in soft-matter viscoelasticity, black-hole horizon physics, Hawking-radiation cutoff, entanglement-bandwidth modulation, and Q-Compute platform-coherence saturation — six domains, one substrate rule-type, no contradiction.

Honest framing of the cross-domain claim. Whether V5 is the *same kernel* (with parameters substrate-derived from a single substrate-cosmology coupling) or the *same rule-type* (with parameters domain-set by separate empirical matching in each arc) is the load-bearing distinction. The first interpretation requires Route A in Paper ED-SC 4.2 (substrate-derived $\ell_{V5}(H_0)$) to close. **As of 2026-05-17 Route A is closed at substrate-parameter-INHERITED level (Paper_027-analog)** — $H_0 = (8\pi/3)^{1/2} \ell_P c^{1/2} \Theta_{ED}^{1/2}$, with $\Theta_{ED}^{\text{Planck-units}} \approx 10^{-122}$ inherited at the P12 primitive level — so the V5 cross-domain pattern is now “one kernel at substrate-parameter-INHERITED level,” not merely “one rule-type with un-derived parameter relations.” This is the same tier of closure Paper_027 gives Newton’s G (form derived; value inherited via one substrate parameter); it is **not** a from-sub-primitives derivation of Θ_{ED} itself, which remains substrate-research-frontier (Routes B/C, the M1 upgrade). Consistent with §7 (SCBU) and §8 (ED-SC 4.x).

§4 — The ED Architecture

The ED program has a four-layer architecture:

Layer 1 — Substrate primitives (P01–P13). Posited ontology. The thirteen primitives.

Layer 2 — Kernel calculus (V1 / V5). Substrate dynamics. The two kernels, classified by characteristic scale, retardation order, and cross-chain rank.

Layer 3 — Domain arcs. Closed research arcs deriving specific physical structures from primitives + kernels + paper-specific postulates. Each arc is structurally coherent within itself and integrated with the substrate.

Layer 4 — Cross-scale synthesis. The Scale Correspondence (SC) program and SCBU unification, integrating multiple arcs into a unified cross-scale invariance model.

Conceptually:

Layer 1: 13 Primitives (P01-P13)

Layer 2: Kernel Calculus (V1, V5)

QM Arc	QFT Arc	Gravity ...	Soft Matter
Phase-1	T17/T18	Arc	/ NS-MHD

Layer 4: Cross-Scale Synthesis (SC, SCBU)

The architecture is **bottom-up**: every result in Layers 3 and 4 traces back through the kernel calculus to the substrate primitives. The architecture is also **falsifiable at each layer**: refutation of a primitive falsifies downstream results; refutation of a kernel-level claim falsifies dependent arc results; refutation of an arc result limits but does not destroy the substrate.

This layered falsifiability is the program’s structural integrity.

§5 — The Domain Arcs

The ED corpus is organized into eight **structurally complete research arcs with declared open derivations**. Each arc:

- States its substrate-level inputs explicitly.
- Identifies its paper-specific postulates (declared, not derived).
- Derives or composes its structural content.
- Distinguishes form-forced content from value-inherited content.
- Provides explicit falsification criteria.

The eight arcs:

Quantum Kinematics (Papers 001–012). The Phase-1 program. Sixteen theorems closing the four standard quantum-mechanical postulates as substrate-derived results. Born rule, tensor-product structure, projective measurement, unitary evolution: all derived from substrate primitives. Berry phase, Aharonov–Bohm phase, Bloch theorem, local rate of becoming.

Quantum Field Theory (Papers 013–024). Form-level QFT scaffolding. T17 identifies gauge fields as connections on rule-type bundles. T18 establishes the kernel-level arrow of time. The Yang–Mills arc (YM-1 through YM-6) supplies substrate-level non-abelian gauge theory, an OS-positivity audit, and a mass-gap mechanism. **Verdict register, honestly: a substrate-level mass-gap mechanism is identified, with explicit paper-specific postulates declared (P-Gap-Coercivity, P-Profile-Rescaling, P-Quartic-Sign). A closed-proof reduction to the Clay-problem mass-gap statement is *not* claimed.** Per Paper_095 methodology, the verdict tier is M2 (Intermediate Path C): structural mechanism identified, postulates declared, constructive proof OPEN.

Gravity (Papers 025–038). Substrate-level identifications of Newton’s law structural form ($G = c^3 \ell_{\text{ED}}^2 / \hbar$, with ℓ_{ED} fixed to the Planck length ℓ_P by **Newton-recovery — a value-inheritance step, not a first-principles derivation of G**); the transition acceleration $a_0 = cH_0/(2\pi)$ as a structural projection of the cosmic decoupling surface; the ED Combination Rule $a = \sqrt{a_N a_0}$; and the BTFR slope-4 relation $v^4 = GMa_0$. The **BTFR slope-4 with zero intrinsic scatter in deep-MOND** is the gravity arc’s sharpest near-term falsifier; current SPARC-class data (McGaugh-Lelli-Schombert 2016: 175 galaxies, slope 3.95 ± 0.08 , scatter ~ 0.1 dex) is consistent at the deep-MOND asymptotic level, with the residual scatter likely

observational rather than intrinsic. Arc ED-10 extends to scalar-tensor acoustic-metric covariantization. The **post-pivot curvature-emergence line** (GR-I ... GR-IV, KM-I/II, One-Field — see the dedicated subsection below) sharpens this Arc-3 content into the weak-field Einstein metric, the khronometric class, and the preferred-frame closure.

Honest scope of the “no dark matter” claim. Substrate gravity explains the **deep-MOND asymptotic** without dark matter: the BTFR slope-4 result with the empirically-consistent $a_0 = cH_0/(2\pi)$ is the load-bearing demonstration. The **transition regime** ($a \sim a_0$, intermediate-acceleration galactic radii) is *predicted* by the ECR $a \approx a_N + a_0 + \sqrt{a_N a_0}$ (Paper_030), but **the corpus does not currently contain explicit per-galaxy SPARC-class residual fits across the transition radius** — those would be the natural next empirical-test program target. The “explains galactic dynamics without dark matter” claim is therefore qualified: *the deep-MOND endpoint is empirically accounted for and the transition regime is structurally predicted; explicit transition-regime residual fits are part of the ongoing empirical-test program (Paper_101 §3.3 register)*. The Whitepaper does not claim the corpus has *already performed* a SPARC-class transition-regime fit; it claims the ECR predicts the transition-regime behavior and BTFR is consistent at the asymptotic endpoint.

Black Holes (Papers 039–052). The Arc Hawking program. Horizon as decoupling surface ($\Gamma_{\text{cross}} \rightarrow 0$). Hawking spectrum at $T_H = \kappa/(2\pi)$ via substrate-Unruh / KMS. Trans-Planckian problem resolved by V5 cutoff at $\omega_c = c/\ell_P$. Planck-mass remnant ($M_\star = c_\star \ell_P$) as forced endpoint (Scenario C). Information paradox not generated at substrate level. Important framing: Planck-mass remnants are **not** a dark-matter candidate — galactic dynamics are explained by substrate gravity, not remnants.

Q-Compute (Papers 053–062). The substrate origin of quantum-computing limits. The multiplicity-cap function $\mathcal{M}_{\text{cap}}$ as one substrate object with three projections (Class A engineered-low-multiplicity, Class B global-geometric-rigidity, Class C high-multiplicity-redundancy). Cross-platform unification via critical multiplicity $\mathcal{M}_{\text{crit}}$: matter-wave quantum-classical boundary (140–250 kDa) and qubit-array walls are the same substrate phenomenon. Decoherence-centric $\rightarrow$ multiplicity-centric reframing.

Entanglement (Papers 063–072). Bipartite entanglement architecture substrate-derived. Tensor product, Schmidt decomposition, monogamy (via V5 cross-chain bandwidth), no-signaling (over-determined by three substrate locks), von Neumann entropy via Shannon–Khinchin axioms at substrate level, Bell–Tsirelson polytope. Interpretive capstone: entanglement is the unresolved regime of participation-rule individuation.

Soft Matter / Navier–Stokes (Papers 073–086). The substrate-level NS / MHD program. Paper_073 (DCGT) at verdict tier **M3 within A $\rightarrow$ regime hydrodynamic-window** supplies the substrate $\rightarrow$ continuum bridge (see §1 framing); the remaining arc papers apply DCGT to specific substrate-content. NS-1 dimensional forcing to $D = 3 + 1$. NS-2 substrate $\rightarrow$ continuum coarse-graining. NS-Smoothness at **M2 (Intermediate Path C)** verdict (mass-gap-class structural-positive; closed proof OPEN). NS-MHD H1/H2/H3 closure. Krieger–Dougherty + Maxwell viscoelastic. Canonical NS-Q operating point with $Q \approx 3.5$. Universal Mobility Law. Vortex-stretching obstruction at substrate level.

Wedges / Kernel Theory (Papers 087–101). Foundational substrate-level theorems. The 13-primitive position paper (Paper 087). V1 and V5 canonical references. Memory-kernel cascade across scales. Kernel hierarchy unification. T18 kernel-level arrow of time. Forward-causal primitive structure. Form-FORCED / value-INHERITED methodology formalized. Cross-scale invariance (ED-SC 3.x). Three-regime renormalization-group / 0.6 problem. NS synthesis. Five-sector program overview. Falsification register.

Each arc is a complete research program in its own right. Each composes with the others

through the substrate primitives and the kernel calculus.

Post-Pivot Gravity Line and the Substrate-Evaluation Wave (June 2026)

Two developments postdate the eight-arc numbered structure above and use a distinct naming convention (no numbered slots consumed).

The post-pivot gravity line (curvature emergence). Sharpening — not superseding — the Arc-3 scalar-tensor / MOND papers, a six-paper line derives gravity from the substrate’s irreversible arrow placed *in the law*:

- **GR-I (*The Bandwidth Lapse and the Factor of Two*).** The substrate bandwidth field supplies an emergent metric ($g \sim 1/b$) and, with the commitment-rate law, the temporal lapse ($N^2 \sim b$) — yielding the weak-field Einstein metric, the Schwarzschild relation, the three classical tests, and the **factor-of-two light bending** (ray-traced 2.09 vs predicted 2.00, conformal control at 0) — the historical discriminator that killed scalar gravity.
- **GR-II (*The Arrow’s Fingerprint*).** The gravitational class is **khronometric** — Einstein plus exactly one scalar, the arrow made dynamical — with one causal cone, so $c_T = c$ structurally (the GW170817 constraint as identity) and Lorentz violation universal rather than species-dependent (invisible to the 10^{-20} matter-sector bounds).
- **GR-III (*The Arrow’s Engine*).** The dynamical rule (the commitment-rate law) and the closure of GR-I’s labeled weak-field assumptions.
- **GR-IV (*The Arrow’s Alibi*).** The preferred-frame parameter computed — $\alpha_1 = -4\lambda_{\text{local}}$, $\lambda_{\text{local}} \sim \rho_{\text{event}}/\rho_{\text{Planck}}$ — and **suppressed below the bounds by more than 70 orders of magnitude** because commitment is sparse (forced by the quantum sector: dense commitment is the Zeno limit, which would abolish quantum mechanics). The preferred-frame front — long the most likely killer of any theory of this class — is closed favorably, and remains a live test ($\alpha_1 \gtrsim 10^{-5}$ in any sub-Planck environment kills it).
- **KM-I / KM-II (*The Arrow’s Deep Field / The Arrow’s Horizon*).** The MOND phenomenology as the khronon’s deep-infrared regime (lensing-correct, no added vector field, solar-system screened), and the cosmological sector (dust-clustering dial, one Λ).
- **One-Field (capstone letter).** Gravity, dark matter, dark energy, and the arrow as one field with one scale — *five roles, three ways to kill it, one number to miss*.

Verdict register: the weak-field Einstein metric, the khronometric class, and the α_1 suppression are **derived / identified**; the full **nonlinear** Einstein field equation remains open (Phase-3 GR locates the obstruction but does not cross it — §11).

The substrate-evaluation wave (build-and-run, could-say-no tests). A distinct register: results obtained by *running* the certified substrate rule and reporting what it does — including where it says *no*.

- **A1 — Common Cause, Not Channel.** Controlled channel capacity across the determinability boundary is *exactly zero*; the boundary is an observational common-cause structure, not a transmission channel.
- **B4 — Charge as Topology.** Electric charge as quantized winding with an integral Gauss law — the topological *skeleton* of charge realized, the metrical *flesh* (the determined Coulomb field) withheld by construction.
- **The Continuum is a Kinetic Lattice-Gas, not a PDE.** The certified substrate’s continuum limit is a ballistic lattice-gas; a closed diffusion PDE appears only when the substrate’s own arrow-selection (Σ) is deleted — “you reach the PDE only by leaving ED.”
- **Primes — the Finite-Memory Ceiling.** Finite-memory ED reproduces the prime *template* (~ 1.700 bits, exact, scale-invariant) but provably **cannot** produce the *escape*

layer (primality / the parity barrier). The program's only *proven* negative.

- **Form and Flesh — The Two Walls.** The reach statement: Wall 1 (coarse-graining — the continuum shadow, mostly open) and Wall 2 (finite-memory — proven, the prime ceiling). What ED supplies is *form*; what it withholds is *flesh*; the non-transfer between the walls is load-bearing.

This register is the program's strongest honesty credential: a substrate that is *run*, not merely argued, and that is explicit about its own edges. (A recent cosmology addition, Paper_ED_CCC, identifies the structure of Penrose's Conformal Cyclic Cosmology within the substrate and supplies two premises CCC assumes.)

§6 — Scale Correspondence (SC) Program

The Scale Correspondence (SC) program is ED's account of how substrate-level structure manifests at the scales accessible to physical experiment.

Why SC is needed. The substrate operates at the Planck scale ($\ell_P \sim 10^{-35}$ m). Physical experiments operate at scales from atomic to cosmological. The gap is sixty-plus orders of magnitude. For ED's substrate to make contact with physics, a *scale correspondence* between substrate content and experimentally-accessible content must be identified.

SC versus Wedges/RG. The Wedges arc (Papers 087–097) establishes the substrate's renormalization-group structure: three regimes (ultraviolet / transition / infrared) with characteristic 0.6 transition exponent, cross-scale invariance at the canonical operating point $\xi_{\text{canonical}}$, and the kernel cascade across scales. This is the **internal substrate-side** structure of cross-scale relations.

The SC program is the **external** counterpart: how does substrate-level structure project onto physically-observable scales? Where do the experimental signatures live? Why does a_0 have the value it has? Why does the matter-wave quantum-classical boundary sit at 140–250 kDa?

Role of SCBU. The Substrate–Cosmology Boundary Unification supplies the central anchor for the SC program. By identifying the substrate-cosmology boundary as a single shared object whose multiple projections supply the canonical scales of physics, SCBU provides the substrate-level *reason* why specific empirical scales exist where they do.

§7 — SCBU: Substrate–Cosmology Boundary Unification

What SCBU claims, with honest scope (post-2026-05-17 Route A closure). SCBU is **M3-supported structural claim at substrate-parameter-INHERITED level**: the substrate-cosmology boundary at $R_H = c/H_0$ — where V5 cross-chain bandwidth saturates — is a single substrate-level object, derived substrate-graph at substrate-parameter-INHERITED level (Paper_027-analog) via the Route A audit cascade. Multiple physical quantities at different domains are projections of this single object onto different parameter axes, now M3-supported by the Cosmology Arc M3 trinity + ED-SC 4.x arc-wide upgrade.

Route A closure delivers the load-bearing derivation that previously stood as the open question: substrate-derived $\ell_{V5}(H_0)$ relation between V5 bandwidth and the cosmological rate, closed at substrate-parameter-INHERITED level. $H_0 = (8\pi/3)^{1/2} \ell_P c^{1/2} \Theta_{\text{ED}}^{1/2}$ with $\Theta_{\text{ED}}^{\text{Planck-units}} \approx 10^{-122}$ inherited at P12 primitive level. The V5 kernel performs structurally identical *jobs* across domains (Maxwell relaxation at $\tau_M \sim 10^{-3}$ s in soft-matter, cutoff $\omega_c = c/\ell_P$ in Hawking,

entanglement-bandwidth modulation, Q-Compute multiplicity cap, NS-Q dissipation, BH decoupling) — and per the Route A closure, these jobs are now M3-supported as performed by *the same* kernel with parameters substrate-graph-derived at substrate-parameter-INHERITED level. The “same kernel vs same rule-type” load-bearing distinction is now closed in favor of “same kernel at substrate-parameter-INHERITED level”; the honest framing in §1 reflects this.

The six projections identified in the corpus (counting Paper_ED_SC_4.1’s two parameter-axis arms — r_H at local-gravitational mass and R_H at global cosmological rate — as two separate projections):

Projection	Parameter axis	Substrate identity
Black-hole horizon $r_H = 2GM/c^2$	local gravitational mass M	cosmological-regime, local-grav arm
Cosmic decoupling $R_H = c/H_0$	global cosmological rate H_0	cosmological-regime, global arm
Substrate fixed-point $\xi_{\text{canonical}}$	substrate-scale fixed-point	spans IR-to-cosmological
Deep-MOND acceleration $a_0 = cH_0/(2\pi)$	local galactic acceleration	IR-galactic projection
Q-Compute threshold $\mathcal{M}_{\text{crit}}$	multiplicity-budget	IR-platform projection
NS-Q canonical $Q \approx 3.5$	hydrodynamic Q-factor	IR-hydrodynamic projection

The cosmological regime (regime 4) carries two arms; the IR regime (regime 3) hosts three sub-projection manifestations. Together these form the canonical six-projection view of the unified four-regime SCBU model (per Paper_ED_SC_4.6).

Why SCBU is offered. Each of these quantities was, prior to SCBU, an independent feature: a_0 derived in the gravity arc, $\mathcal{M}_{\text{crit}}$ in the Q-Compute arc, Q in the soft-matter arc. They share a striking common feature: each is structurally anchored to the cosmological rate H_0 through the substrate-cosmology boundary, even though each lives in a domain (galactic dynamics / quantum-computing / hydrodynamics) that has no operational connection to cosmology. SCBU proposes that this is not coincidence: that from the substrate’s perspective, these are the same boundary observed via different projection mechanisms.

What SCBU does not yet do. SCBU does not currently derive the specific scaling exponents per projection axis. “Co-varies with H_0 ” admits many functional forms. The prediction sharpens dramatically if SCBU forces specific exponents (e.g., $a_0 \propto H_0^1$ — already structural in Paper_029 — but $\mathcal{M}_{\text{crit}} \propto H_0^\alpha$ with α fixed by substrate structure has not been derived). Until per-projection exponents are fixed, the synchronized-covariation prediction is a *family of falsifiers* indexed by free exponents, rather than a single sharp falsifier with no degrees of freedom. The §9 falsification register reflects this: BTFR slope-4 (Paper 031) is the corpus’s sharpest empirically-clean falsifier; SCBU’s covariation prediction is its most structurally far-reaching falsifier but currently softer.

The structural claim, honestly framed (post-Route-A-closure): *one substrate, one boundary, six projections — M3-supported structural claim at substrate-parameter-INHERITED level (Paper_027-analog).* Substrate-cosmology coupling (Route A) closed at substrate-parameter-INHERITED level per multi-route convergence audit option (ii); $\Theta_{\text{ED}}^{\text{Planck-units}} \approx 10^{-122}$ inherited at P12 primitive level. The strength of the claim is now M3 across all six projections; tightening for per-projection exponent derivation is substrate-research-frontier work (not load-bearing). The claim shrinks if the empirical covariation prediction is falsified along any single projection axis.

§8 — The ED-SC 4.x Arc

The ED-SC 4.x arc is a six-paper extension that formalizes the SCBU framework’s six projections and integrates them into a unified four-regime cross-scale model.

Paper ED-SC 4.1 — BH cosmic boundary projection. Establishes that the black-hole horizon r_H and the cosmic decoupling surface R_H are two arms of one substrate-cosmology boundary, parameterized by local mass M and global cosmological rate H_0 respectively. Two distinct projection mechanisms are identified: the BH-horizon arm uses a cross-scale fixed-point + Schwarzschild matching, while the cosmic decoupling arm uses Paper_028’s substrate-causality / decoupling-surface argument (kernel-propagation-rate / Hubble-expansion-rate crossover). Both arms share the substrate-cosmology boundary anchor but employ different projection mechanisms.

Paper ED-SC 4.2 — Substrate-derivation of $\xi_{\text{canonical}}(H_0)$ (post-Route-A-closure). The highest-leverage paper. Provides first-principles substrate derivation of the canonical operating point as a function of H_0 . *Route A closed at substrate-parameter-INHERITED level (2026-05-17) via the Route A audit cascade.* The structural setup — $\xi_{\text{canonical}}$ as a fixed point of substrate renormalization-group flow on the interval $[\ell_P, R_H]$ — is now closed at substrate-parameter-INHERITED level (Paper_027-analog): $\xi_{\text{canonical}} \sim \sqrt{\Theta_{\text{ED}}^{\text{Planck-units}}} \cdot M_P$ via Route A4 chain. Routes B and C carry substrate-research-frontier tightening status. The honest outcome: structural framework supplied; numerical closure at substrate-parameter-INHERITED level; per-projection exponents derivable in principle via Route A inheritance chain.

Paper ED-SC 4.3 — MOND/galactic IR projection. Establishes the deep-MOND acceleration scale $a_0 = cH_0/(2\pi)$ as the galactic-acceleration sub-projection of the substrate-cosmology boundary, with the factor $1/(2\pi)$ arising from Paper_029’s azimuthal-Fourier-mode normalization on the residual SO(2) symmetry of an accelerating chain. This is a **distinct** projection mechanism from R_H ’s substrate-causality argument (4.1) — a_0 and R_H share the boundary anchor but use different mechanisms onto different parameter axes. BTFR slope-4 emerges as the infrared-regime scaling law at the boundary.

Paper ED-SC 4.4 — Q-Compute SCBU IR projection. Establishes $\mathcal{M}_{\text{crit}}$ and the Class A/B/C saturation structure as the multiplicity-budget projection of the boundary. The three architectural classes correspond to which of the three substrate constituents ($N_{\text{bw}}, N_{\text{V5}}, N_{\text{commit}}$) saturates first as the boundary is approached. Cross-platform universality (matter-wave at 140–250 kDa, qubit-array walls) inherits from the substrate’s structural invariance.

Paper ED-SC 4.5 — Soft-matter NS-Q IR projection. Establishes $Q \approx 3.5$ at the canonical NS-Q operating point as the hydrodynamic-Q projection of the boundary. V5’s appearance as the Maxwell relaxation timescale ($\tau_M \sim \tau_{\text{V5}}$) shares the substrate object that saturates at the boundary. The Universal Mobility Law inherits from the same substrate kernel hierarchy.

Paper ED-SC 4.6 — Unified four-regime model (capstone). Integrates all five prior extensions. The substrate-cosmology boundary induces a unified four-regime renormalization-group structure: ultraviolet (V1-dominated), transition (V1 V5 cascade with 0.6 exponent), infrared (V5-modulated, multi-sector projection regime), cosmological (V5-saturated boundary). All six SCBU projections occupy distinct, predictable positions within this structure.

What the arc accomplished (post-Route-A-closure, 2026-05-17). The ED-SC 4.x arc completes ED’s cross-scale synthesis. The substrate-cosmology boundary’s structure is mapped across domains; the regime classification is portable; the framework supplies multiple testable predictions (see §9); and **all six projections close at M3 with substrate-parameter-INHERITED foundation via Route A4 chain.** Route A’s closure at substrate-parameter-INHERITED level (Paper_027-analog) simultaneously upgraded the entire arc to M3

form-IDENTIFIED + value-INHERITED + saturation-regime scale DERIVED at substrate-parameter-INHERITED level — **the first cross-arc M3 cosmology result in the corpus** delivered via the Cosmology Arc M3 trinity + ED-SC 4.x arc-wide upgrade.

§9 — Falsification Program

ED is a falsifiable program. Each closed result carries explicit falsification criteria, and the program-level synthesis carries cross-arc tests.

Observational falsifiers (corpus-wide):

- *Direct detection of substrate c variation.* P-RB-1 (Paper 012) commits to substrate c being constant. Detection of fundamental-constant drift in c that cannot be accounted for as a dressed-object coarse-graining effect refutes the framework.
- *Empirical failure of BTFR slope-4* (Paper 031). Substrate gravity predicts the Baryonic Tully–Fisher Relation with slope exactly 4 and zero intrinsic scatter in the deep-MOND asymptotic.
- *Empirical detection of complete black-hole evaporation* (Paper 041). Substrate dynamics force a Planck-mass remnant endpoint. Complete evaporation refutes the V5-cutoff mechanism.
- *Empirical detection of intrinsic scatter in BTFR beyond observational error.* Refutes Paper 031 and propagates upward.

Structural falsifiers:

- *Discovery of a fourth Q -Compute architectural class beyond $A/B/C$* (Paper 055). Refutes the three-constituent exhaustiveness of $\mathcal{M}_{\text{cap}}$.
- *Demonstration that V5 admits trans-Planckian modes.* Refutes Paper 040’s resolution of the trans-Planckian problem.
- *Substrate-level evidence that any of the 13 primitives is inconsistent or derivable from another.* Refutes the minimality claim.
- *Discovery of substrate-level structure below ℓ_P .* Refutes the substrate-interior cutoff (Paper 042).

Cross-arc falsifiers (the SCBU framework’s signature tests):

- *Hubble-tension-class predictions.* SCBU predicts synchronized co-variation of six quantities under H_0 shifts: $\xi_{\text{canonical}}$, a_0 , $\mathcal{M}_{\text{crit}}$, Q , r_H , R_H . **The prediction’s sharpness is currently limited by un-derived per-projection scaling exponents.** $a_0 \propto H_0^1$ is structural (Paper 029); $R_H \propto H_0^{-1}$ is structural (Paper 028); but $\mathcal{M}_{\text{crit}} \propto H_0^\alpha$, $Q \propto H_0^\beta$, and $\xi_{\text{canonical}} \propto H_0^\gamma$ have not been derived with specific exponents. As currently formalized, SCBU predicts “co-variation in some sense” rather than a single-exponent law per projection, which means tensions can be absorbed by adjusting the unmanaged exponents. The prediction is therefore a *family of falsifiers* indexed by per-projection exponents; with Route A now closed (2026-05-17, substrate-parameter-INHERITED level) these exponents are derivable in principle via the Route A inheritance chain, though each specific exponent ($\mathcal{M}_{\text{crit}}$, Q) still requires explicit derivation. **The sharpest test in this register meanwhile is empirical detection of asynchrony** — e.g., a_0 tracking H_0 as predicted while $\mathcal{M}_{\text{crit}}$ remains flat — which would refute the SCBU-projection identification for the asynchronous quantity *regardless* of un-derived exponents.
- *Multi-axis consistency.* Systems parameterizable by multiple SCBU-projection axes (e.g., a high-mass-fluid molecular system at galactic-disc accelerations) should exhibit consistent predictions across all relevant axes.

- *Cross-platform universality.* Different platforms probing the same SCBU projection should yield substrate-mapped agreement. Different platforms probing $\mathcal{M}_{\text{crit}}$ (matter-wave vs qubit-array) must map to the same substrate threshold.

These tests are sharp. They do not require new physics to perform — they require careful observation of existing measurable quantities under cosmologically-varied conditions. The Hubble tension, if it persists, is itself a candidate testing ground.

§10 — How to Read the ED Corpus

The full corpus comprises 100+ research papers (101 numbered + theorem-stub and synthesis additions). A complete reading is not necessary for orientation. The following **core papers** form a coherent introductory path:

Foundational (read first):

1. **Paper 087** — The 13 primitives (position paper). Sets the substrate ontology.
2. **Paper 089** — V1 finite-width retarded kernel (canonical reference). The first kernel.
3. **Paper 090** — V5 cross-chain correlation kernel. The second kernel.
4. **Paper 073** — Diffusion Coarse-Graining Theorem (DCGT). The substrate-to-continuum bridge.
5. **Paper 095** — Form-FORCED / value-INHERITED methodology. The corpus’s methodological discipline formalized.

One paper per major arc (read in any order):

6. **Paper 002** — Born–Gleason rule (quantum kinematics anchor).
7. **Paper 027** — Newton’s $G = c^3 \ell_P^2 / \hbar$ (gravity arc anchor).
8. **Paper 047** — Hawking spectrum via substrate-Unruh (black-hole arc anchor).
9. **Paper 060** — Matter-wave / qubit unification via $\mathcal{M}_{\text{crit}}$ (Q-Compute anchor).
10. **Paper 076** — NS-2 substrate→continuum (soft-matter / NS anchor).

Synthesis (read after core):

11. **Paper 098** — ED-QFT Unified Overview. Program synthesis covering five sectors.
12. **Paper 100** — Five-Sector Program Overview. The 100-paper milestone capstone.
13. **Paper SCBU** — Substrate–Cosmology Boundary Unification.
14. **Paper ED-SC 4.6** — Unified four-regime model (ED-SC 4.x arc capstone).

Post-pivot (the newest front, read for the current state):

15. **One-Field letter** — the post-pivot gravity line’s front door (gravity + dark matter + dark energy + the arrow, one field; entry to GR-I ... GR-IV, KM-I/II).
16. **Form and Flesh: The Two Walls** — the program’s reach statement: what ED supplies (form) and what it provably withholds (flesh), anchored by the finite-memory prime ceiling.

Navigation strategy:

- Each paper begins with a **Preamble** listing what the paper does *not* claim. Read these first when sampling.
- Each paper has a **§2.5 Load-Bearing Step Audit Table** that lists every step in the derivation with its discipline-label (postulate / inherited / derived / position-claim). The audit table is the paper’s structural skeleton.
- Each paper closes with explicit **Falsification Criteria** and a **Position Statement**.

- Cross-references are dense but the corpus is structured to allow targeted reading: enter at an arc anchor, follow citations to dependencies, exit when the dependency chain becomes shallow.

The corpus rewards focused reading. A physicist coming from quantum mechanics can profitably read the quantum-kinematics arc + the entanglement arc + the form-level QFT arc without touching gravity. A cosmologist can read the gravity arc + the BH arc + the SCBU synthesis. A condensed-matter physicist can read the soft-matter arc + the Q-Compute arc without touching cosmology.

§11 — Future Work

Several open lines of work would significantly advance the program:

The substrate-parameter frontier (the M1 upgrade). Route A — the substrate-level $\ell_{V5}(H_0)$ relation — **closed at substrate-parameter-INHERITED level on 2026-05-17** (Paper_027-analog), upgrading the ED-SC 4.x arc to M3 arc-wide and delivering the Cosmology Arc M3 trinity (§8). What remains open is the *fuller* closure: deriving the substrate parameter $\Theta_{ED} \approx 10^{-122}$ (and τ_{V5}) from sub-primitive content — the analog of the longstanding “what fixes the values of $\ell_P, \hbar, c$?” question. Closing this via Route B (substrate-derived $\beta(K)$) and/or Route C (substrate-derived K_{boundary} at R_H), without additional postulates, would upgrade the arc from M3 to **M1 — the first cross-arc M1 result in the corpus**. This is substrate-research-frontier work, not load-bearing for the current M3 verdicts.

Hawking cosmological-redshift extension. A natural next ED-SC 4.x extension paper: substrate-Unruh KMS structure at the SCBU boundary should connect Hawking temperature to cosmological redshift via the same boundary-projection mechanism that connects a_0 and $\mathcal{M}_{\text{crit}}$. This would add a seventh SCBU projection on the Hawking-temperature parameter axis.

Remnant cosmological-relic-abundance extension. Paper 049 introduces the Planck-mass remnant relic abundance under standard FRW dilution. An explicit SCBU framing would tighten the structural anchoring of remnants to the substrate-cosmology boundary.

Entanglement V5-saturation SCBU. Paper 071 establishes an ER=EPR-class structural echo between V5 in the entanglement arc and V5 in the BH arc. An explicit SCBU framing would integrate the entanglement arc into the cross-scale synthesis.

Yang–Mills SCBU. The deepest open connection. Yang–Mills mass-gap mechanism (Paper 021) and OS-positivity audit (Paper 020) both invoke V5 at substrate level. An explicit SCBU framing would integrate the YM arc into the unified four-regime model, conjecturally as a tenth projection.

Phase-3 GR: the full Einstein equation. The post-pivot gravity line (§5) now derives the **weak-field** Einstein metric (GR-I, the factor of two), pins the gravitational class (GR-II, khronometric), closes the preferred-frame front (GR-IV, α_1 suppressed by more than 70 orders), and unifies dark matter / dark energy / the arrow (KM-I/II, One-Field). What remains open is the **full nonlinear** Einstein field equation: the Phase-3 GR analysis *locates* the obstruction (a fixed-geometry invariant — the same gate as the A2 decoupling and B4 charge results) but does not cross it. Whether the full EFE emerges — and at what verdict level — remains genuinely open; the weak-field, the class, and the preferred-frame safety no longer are.

Empirical program. Observational programs targeting the SCBU synchronized co-variation prediction under Hubble tension; precision galactic-rotation-curve measurements of BTFR scat-

ter; matter-wave decoherence experiments at the 140–250 kDa boundary; substrate-cutoff signatures in Hawking-radiation analog systems.

§12 — Summary

Event Density (ED) is a generative substrate ontology for cross-scale physics built on:

- **One substrate** — the thirteen primitives.
- **Two kernels** — V1 (finite-width retarded) and V5 (cross-chain finite-memory).
- **Eight structurally complete research arcs (with declared open derivations)** spanning quantum mechanics, gauge theory, gravity, black holes, quantum computing limits, entanglement, soft matter, and foundational kernel theory.
- **One boundary** — the substrate-cosmology boundary at $R_H = c/H_0$.
- **Six projections** of that boundary onto distinct parameter axes (black-hole horizon, cosmic decoupling, substrate fixed-point, deep-MOND acceleration, Q-Compute threshold, hydrodynamic Q-factor).
- **Four regimes** of the unified renormalization-group structure (ultraviolet, transition, infrared multi-sector, cosmological).

The program’s methodological signature: *form forced, values inherited*. Structural content of physical results is derived from the substrate ontology and explicit paper-specific postulates; numerical values are inherited from empirical matching. Every load-bearing step is labeled, every falsification path is explicit, every open question is flagged.

Two June-2026 developments extend the picture beyond the eight arcs: a **post-pivot gravity line** (GR-I ... GR-IV, KM-I/II, One-Field) deriving the weak-field Einstein metric, the khronometric class, the preferred-frame safety, and the MOND / dark-energy unification from the arrow placed in the law (the full nonlinear Einstein equation still open); and a **substrate-evaluation wave** of build-and-run tests — including the program’s only *proven* negative (the finite-memory prime ceiling) and its reach statement, *Form and Flesh: The Two Walls*.

The program’s most ambitious **hypothesis**: that the substrate-cosmology boundary is one substrate-level object, and the apparent independence of its multiple physical manifestations — across gravity, quantum computing, soft matter, cosmology — is an artifact of viewing it through domain-specific parameter axes. The unified four-regime model offers a schema for this hypothesis. Whether it closes as a derivation rather than an organizing schema turned on Route A; **as of 2026-05-17 Route A is closed at substrate-parameter-INHERITED level** (Paper_027-analog, per Paper ED-SC 4.2 §1.5), making the unified picture an M3-supported structural claim. The remaining frontier is deriving the substrate parameter Θ_{ED} itself (Routes B/C $\rightarrow$ M1).

The program’s **sharpest currently-clean falsifier**: BTFR slope-4 with zero intrinsic scatter in deep-MOND (Paper 031). Testable against present galaxy-rotation-curve catalogs without requiring SCBU closure.

The program’s **most structurally far-reaching** (but currently softer) test: synchronized co-variation of the six projections under Hubble-tension-class H_0 shifts. Sharpness here grows with per-projection scaling-exponent derivations, which are part of the Route A program.

The program is unfinished. The substrate-parameter derivation (Θ_{ED} ; Routes B/C $\rightarrow$ M1) is open. Phase-3 general-relativity emergence is open. Empirical confirmations of the SCBU prediction structure are open. Yet the cross-domain reach across eight arcs, combined with the structural honesty of the methodology, is the program’s academic case.

ED is a research program, not a final theory. It proposes a substrate-level account of physics that is falsifiable at every layer, integrated across every arc, and synthesized at the cross-scale level. The whitepaper is the front door; the 100+ paper corpus is the building behind it.

End of Whitepaper.

For navigation to the corpus, see PAPERS_INDEX. For the primitive enumeration, see paper_ED_Framework_13_Primitive_Generative_System.md. For corpus-level dependency structure, see DEPENDENCY_GRAPH_ED.md. For falsification register, see Paper 101.